PADHAKU
Exploration · Class 9 Science



CHAPTER 2

Cell: The Building Block of Life

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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🤔 Think It Over

1 **Where does a cell come from?**

✔ ANSWER

A new cell always comes from a **pre-existing cell** by cell division (Rudolf Virchow, Cell Theory).

The first cells probably formed in water (for example, small water pools like hot springs), when a protective membrane formed around organic molecules.

2 **How have technological interventions facilitated the creation of new knowledge in understanding the world beyond the naked eye?**

✔ ANSWER

Our eye cannot separate two points closer than **0.1 mm**, so cells cannot be seen without help.

- Robert Hooke's microscope (1665) showed cells in cork for the first time.
- Light microscopes with better lenses improved magnification, resolution and contrast.
- Electron microscopes use a beam of electrons and show cell organelles at the nanometre scale.

So each new instrument revealed new details and created new knowledge about cells.

3 How is the cell the structural and functional unit of life?

✔ ANSWER

- **Structural:** every organism is made of one or more cells; cells form tissues, tissues form organs and organs form organ systems.
- **Functional:** all life activities (getting energy, making materials, removing waste, growth, reproduction) take place inside cells.

4 How does a cell multiply?

✔ ANSWER

By **cell division**: one parent cell divides to form new daughter cells.

- **Mitosis** forms 2 identical daughter cells (growth, repair).
- **Meiosis** forms 4 gametes with half the chromosomes (sexual reproduction).

⭐ 2.1 How to Study Cells?

? **What do we call the ability of the human eye to see two very close objects as separate and distinct?**

✔ ANSWER

It is called **resolution** (resolving power). The **limit of resolution** of the human eye is **0.1 mm** (seen from about 25 cm).

Activity 2.1 **Let us estimate the size of a cell: field of view diameter = 5 mm, 25 onion cells along the diameter. Estimate the cell size. If eyepiece and objective are both 10X, how much does the microscope magnify the cell?**

✔ ANSWER

Given: diameter of field = 5 mm = 5 × 1000 = 5000 µm; number of cells = 25

Formula: size of one cell = diameter of field (µm) ÷ number of cells

Size = 5000 µm ÷ 25 = **200 µm**

Total magnification = eyepiece × objective = 10X × 10X = **100X**

So the 200 µm cell appears 100 times larger.

2.2.1 Cell membrane — The universal feature of a cell

? **How does the structure of the cell membrane in the cells of alveoli control the movement of substances across it?**

✔ ANSWER

The membrane of alveoli cells is very thin and **selectively permeable** (lipid bilayer with proteins).

Small gas molecules like oxygen and carbon dioxide pass through it easily by **diffusion**, from higher to lower concentration: oxygen into the blood, carbon dioxide out of it. Larger or unwanted substances are blocked.

Activity 2.2 **Potato pieces of equal size: one in plain water (Beaker A), one in 20 per cent salt or sugar solution (Beaker B). Weigh before and after about an hour. What do you observe and infer?**

✔ ANSWER

Observation: Beaker A – the piece swells and its weight increases. Beaker B – the piece shrinks and its weight decreases.

Conclusion: The cell membrane lets water pass but not salt or sugar. Water moves from a dilute to a concentrated solution through a selectively permeable membrane. This is **osmosis**: water enters the cells in A and leaves the cells in B.

What if... **What if mung bean seeds are kept in a concentrated solution after soaking in water for 12 hours? What will happen to them?**

💡 SAMPLE ANSWER

The solution outside is more concentrated, so water moves **out** of the seed cells by osmosis. The seeds lose water, shrink and wrinkle, and may not germinate.

What if... **What if a cell is kept in salt or sugar solutions of different concentrations?**

✔ ANSWER

- **Isotonic** (outside = inside concentration): no net water movement, the cell stays the same.
- **Hypotonic** (outside < inside): water enters, the cell swells (an animal cell may burst).
- **Hypertonic** (outside > inside): water leaves, the cell shrinks.

? **Cells of plants, fungi and bacteria have a cell wall. What do you think is the necessity of the cell wall in these cells?**

✔ ANSWER

The cell wall is a rigid outer layer. It gives the cell a fixed shape, strength and protection, stops it from bursting when water enters, and (in plants) helps the plant stand upright against wind and rain.

Activity 2.3 **Observe onion peel (or Rhoeo) cells and human cheek cells under a microscope. Why are they different? What happens when 20 per cent sugar solution is put on them?**

✔ ANSWER

Observation: onion/Rhoeo cells are box-shaped and regularly arranged; cheek cells are irregular in shape and loosely arranged.

Reason: plant cells have a rigid **cell wall** that gives a fixed shape; cheek cells have only a flexible cell membrane.

In sugar solution: in Rhoeo cells the outer boundary (cell wall) stays the same but the inner content shrinks away from the wall; cheek cells shrink as a whole. Both lose water by osmosis, but the cell wall keeps the plant cell's shape.

⭐ Pause and Ponder · 2.2.2 Cell wall

1 **What argument would you give for the necessity of a cell wall in plants usually fixed in one place versus in animals usually moving from one place to the other?**

✔ ANSWER

Plants cannot move away from wind, rain and other stresses, so they need a **rigid cell wall** for support, a fixed shape and to stay upright.

Animals move from place to place, so their cells need to be **flexible** and change shape; a rigid wall would stop this movement. Animals get support from skeletons instead.

2 **What consequences would you predict for a plant cell if its cell wall were to become as flexible as a cell membrane?**

💡 SAMPLE ANSWER

- The cell would lose its fixed shape and firmness.
- In water it could swell too much and might burst; in a concentrated solution it would shrink fully.
- Leaves, flowers and stems would droop and the plant could not stand upright.

3 **Why is it important to cut the two potato pieces in roughly equal size and measure**

their initial weight before placing them in different liquids?

✔ ANSWER

Equal size makes it a **fair test**: the only difference is the liquid.

The initial weight is needed to calculate the **change in weight** (final – initial), which shows whether water entered or left the cells.

⭐ 2.3 The Cell Interior — A Coordinated Working System

Activity 2.4 **Study the bacterial, plant and animal cells (Fig. 2.10) and complete Table 2.1.**

✔ ANSWER

Cell structure	Bacterial cell	Plant cell	Animal cell
Cell membrane	Present	Present	Present
Cell wall	Present	Present	Absent
Cytoplasm	Present	Present	Present
Well-defined nucleus	Absent	Present	Present
Nucleoid (primitive nucleus)	Present	Absent	Absent
Membrane-bound organelles	Absent	Present	Present

? **Which of the cells given in Fig. 2.10 fall under the categories of prokaryotic and eukaryotic**

cells?

✔ ANSWER

- **Prokaryotic:** bacterial cell (Fig. 2.10a) – no well-defined nucleus, no membrane-bound organelles.
- **Eukaryotic:** plant cell (Fig. 2.10b) and animal cell (Fig. 2.10c).

? **Mature red blood cells have no nucleus. Do you know any other cells without nucleus?**

💡 SAMPLE ANSWER

Yes. Mature **sieve tube cells** of phloem in plants have no nucleus. Mature cells in the lens of the human eye also lose their nucleus.

? **How does the cell prevent wastes from accumulating inside it?**

✔ ANSWER

With **lysosomes**: sacs full of enzymes that break down waste, unwanted molecules and worn-out organelles. The products are released into the cytoplasm and reused.

? **Are there any other plastids in plant cells that contain pigments other than the green pigment? How do flowers, fruits and vegetables acquire varied colours?**

✔ ANSWER

Yes, **chromoplasts**. They contain yellow, orange or red pigments, which give bright colours to flowers and fruits (these attract pollinators and seed-dispersing animals).

Colourless plastids are **leucoplasts**; they store starch, oils or proteins.

? **Where are water, minerals and waste materials stored in the cell? Why do plants look wilted when they do not get enough water?**

✔ ANSWER

They are stored in the **vacuole** (filled with cell sap). A mature plant cell has one large central vacuole.

Its water keeps pressure inside the cell, so the cell is firm. Without enough water the vacuole loses water, cells become less firm and the plant **wilts**.

4 **Do white flowers contain any pigment? Give reasons.**

💡 SAMPLE ANSWER

Usually **no coloured pigment**. Their plastids are colourless (leucoplasts), not chromoplasts.

They look white because their cells and the air spaces between them reflect all the colours of light equally.

5 **Draw a well-labelled schematic diagram of a plant or an animal cell using these clues:**

(i) Nucleus appears as a dark and round body inside the cell.

(ii) ER spreads like a network of extended nuclear envelope.

(iii) Mitochondria and chloroplasts are rod shaped.

💡 SAMPLE ANSWER

Draw like Fig. 2.10 (b) (plant cell):

- Outer rectangle = **cell wall**, a line just inside it = **cell membrane**.
- Dark round **nucleus** (with nucleolus) near one side.
- **ER** as a network of lines starting from the nuclear membrane, with dots (ribosomes) on the rough ER.

- Rod-shaped **mitochondria** and rod-shaped green **chloroplasts** in the cytoplasm.
- Stacks of flat sacs = **Golgi apparatus**; a large **vacuole** in the centre.

(For an animal cell: no cell wall, no chloroplast, small vacuoles, add lysosomes.)

⭐ 2.4 How do Normal Cells Grow and Divide?

Activity 2.5 **Onion root tip squash: Do you observe the roots growing? Do you observe the cells of the onion root tip? Are they similar in structure? Do you find any structural differences? If yes, why?**

✔ ANSWER

Observation: new roots grow in 5–6 days. The cells look alike in shape, but their nuclei look different: in some the nucleus is round, in some thread-like chromosomes are seen, in some chromosomes are in the middle or moving apart, and some have two nuclei.

Reason: cells at the root tip divide continuously, so we see cells at **different stages of cell division**.

? **Can you identify which stage comes first during cell division?**

💡 SAMPLE ANSWER

The stage in which the chromatin in the nucleus **condenses into thread-like chromosomes** comes first (it is called prophase; you will learn the stages in higher grades).

⭐ Pause and Ponder · 2.4 Cell division

6 **Instead of many small ones, why does a cell not have a single giant mitochondrion? How does this relate to the concept of surface area?**

✔ ANSWER

Energy production happens on the membranes (cristae), so it needs a large **surface area**.

Many small mitochondria together have a much larger surface area than one giant mitochondrion of the same volume, so they make more energy.

Small mitochondria can also spread throughout the cell to wherever energy is needed.

7 **If the skin cells start dividing by meiosis instead of mitosis, what do you think will happen to a cut on the skin?**

✔ ANSWER

Meiosis gives cells with **half the number of chromosomes** that are not identical to the parent cell.

Such cells cannot work like normal skin cells, so the cut would not heal properly; the new tissue would be abnormal and the body could have problems.

⭐ Revise, Reflect, Refine

1 **Differentiate between the following pairs of terms based on the clues given in parentheses:**
(i) Cell membrane and cell wall (permeability)
(ii) RER and SER (structure)
(iii) Chloroplasts and chromoplasts (pigments)

✔ ANSWER

Pair	First	Second
(i) Permeability	Cell membrane: **selectively permeable** – lets some substances (like water) pass and blocks others	Cell wall: **permeable** – water and dissolved minerals pass freely
(ii) Structure	RER: **ribosomes attached** on its surface, looks rough	SER: **no ribosomes**, looks smooth
(iii) Pigments	Chloroplasts: green pigment **chlorophyll**	Chromoplasts: **yellow, orange or red** pigments (not chlorophyll)

2 **Cell X (animal cell) in pure water swells; Cell Y in concentrated salt solution shrinks. Which statement gives the correct explanation?**

(i) Salt molecules moved into Cell Y, causing it to shrink.

(ii) Water moved into Cell X and more water moved out of Cell Y than the salt solution entered in it.

(iii) Water moved into Cell X and moved out of Cell Y through the cell membrane.

(iv) Solute movement caused osmosis in both cells.

✔ ANSWER

(iii) Water moved into Cell X and moved out of Cell Y through the cell membrane.

Reason: this is osmosis – only water crosses the selectively permeable membrane, from dilute to concentrated side; salt does not move in.

3 **Identify parts (a) to (g) in Fig. 2.20 (a plant cell) and match them with their functions:**

(i) Controlling all the activities of a cell.

(ii) Site of cellular respiration.

(iii) Storage organelle that also provides rigidity to the cell.

(iv) Separates the cell contents from surroundings.

(v) Provides structural rigidity to the cell.

(vi) Packs and stores materials received from ER.

(vii) Helps in manufacturing food.

✔ ANSWER

Label	Part	Function
(a)	Mitochondrion	(ii) Site of cellular respiration
(b)	Nucleus	(i) Controlling all the activities of a cell
(c)	Golgi apparatus	(vi) Packs and stores materials received from ER
(d)	Chloroplast	(vii) Helps in manufacturing food
(e)	Cell wall	(v) Provides structural rigidity
(f)	Cell membrane	(iv) Separates cell contents from surroundings
(g)	Vacuole	(iii) Storage organelle that also gives rigidity

4 **Which option(s) are correctly placed under the categories?**

Option	Present in the plant cells	Absent in the animal cells
(i)	Leucoplast	Cell wall
(ii)	Mitochondria	Ribosome
(iii)	Cell wall	Golgi apparatus

(iv)	Lysosome	Endoplasmic reticulum

✔ ANSWER

(i) only.

Leucoplasts are present in plant cells and the cell wall is absent in animal cells. Ribosomes, Golgi apparatus and ER are all present in animal cells, so (ii), (iii) and (iv) are wrong.

5 **Renu says all parts of plants, even roots, contain plastids. Rohit says plastids are absent in roots because roots are underground and do not photosynthesise. Who is correct? Justify.**

✔ ANSWER

Renu is correct.

Roots do not have chloroplasts, but they have **leucoplasts** (colourless plastids) that store food such as starch. Carrot roots have **chromoplasts** (orange colour). So plastids are present in roots too.

6 **Mitochondria and chloroplasts are two important organelles in a plant cell. Discuss how these two organelles are structurally and functionally similar to each other, and different from each other.**

✔ ANSWER

Similarities:

- Both have **two membranes**.
- Both have their **own DNA and ribosomes** and make some of their own proteins.
- Both deal with **energy** in the cell.

Differences:

Feature	Mitochondria	Chloroplasts
Inner structure	Inner membrane folded into **cristae**	**Stroma** with disc-shaped membranes containing chlorophyll
Pigment	No pigment	Green **chlorophyll**
Function	**Cellular respiration**: break down glucose to release energy (ATP)	**Photosynthesis**: use light energy to make food
Found in	Almost all eukaryotic cells	Only green plant cells (and algae)

7 **Which of the following pairs of cell organelles contains DNA?**

(i) Chloroplasts, Ribosomes

(ii) Mitochondria, Nucleus

(iii) Golgi bodies, Ribosomes

(iv) Nucleus, Lysosomes

✔ ANSWER

(ii) Mitochondria, Nucleus. Both contain DNA; ribosomes, Golgi bodies and lysosomes do not.

8 (i) **A researcher placed one carrot in plain water and a similar carrot in concentrated salt solution, and observed after 24 hours. What hypothesis does she want to test?**

✔ ANSWER

"Water moves into or out of the cells of the carrot by **osmosis**, depending on the concentration of the solution around it" – the carrot gains water in plain water and loses water in salt solution.

8 (ii) **What would you suggest for the improvement of this experiment?**

💡 SAMPLE ANSWER

- Weigh (and measure) both carrots before and after.
- Use carrots/pieces of the same size and the same volume of liquid, at the same temperature.
- Repeat with several carrots and with a few salt concentrations.
- Note observations at regular times, not only after 24 hours.

8 (iii) **Why does the carrot in plain water stay stiff and crunchy, but the carrot in concentrated salt solution become rubbery and limp?**

✔ ANSWER

In plain water, water enters the cells by osmosis; the vacuoles fill up and press against the cell walls, so the cells are firm – the carrot stays **stiff and crunchy**.

In salt solution (more concentrated outside), water leaves the cells by osmosis; the cells lose firmness and the carrot becomes **rubbery and limp**.

9 **Indicate the presence or absence of the following structures in bacterial and animal cells: chromosome, nucleus, mitochondria, Golgi complex, chromoplasts.**

✔ ANSWER

Structure	Bacterial cell	Animal cell
Chromosome	Present (single circular DNA in the nucleoid)	Present
Nucleus	Absent (only nucleoid)	Present
Mitochondria	Absent	Present
Golgi complex	Absent	Present
Chromoplasts	Absent	Absent

10 (i) **Potato cups in beakers of water: Cup A empty; Cup B one teaspoon sugar; Cup C one teaspoon salt; Cup D (boiled potato) one teaspoon sugar. After two hours:**

Explain why water gathers in the hollowed portion of Cup B and Cup C.

✔ ANSWER

Sugar or salt in the cup makes a **concentrated solution** inside. The water in the beaker is more dilute. Water moves from the beaker through the **living potato cells** (selectively permeable membranes) into the cup by **osmosis**.

10 (ii) **Why is Cup A necessary for this experiment?**

✔ ANSWER

Cup A is the **control**. It shows that without sugar or salt no water collects, so the water in Cups B and C is due to the solute.

10 (iii) **Explain why water does not gather in the hollowed portions of Cups A and D.**

✔ ANSWER

- **Cup A:** no solute inside, so there is no concentration difference and no osmosis into the cup.
- **Cup D:** boiling kills the potato cells and damages their membranes, so they are no longer selectively permeable and osmosis does not happen.

11 **Identify the pair that incorrectly matches the cell organelle with its function.**

(i) Ribosome — Protein synthesis

(ii) SER — Lipid and cellulose synthesis

(iii) Lysosome — Digestion of foreign agents

✔ ANSWER

(ii) SER — Lipid and cellulose synthesis. SER makes lipids (fats) and hormones, but not cellulose.

12 **What outcome do you expect, if all the mitochondria are removed from a eukaryotic cell?**

✔ ANSWER

Cellular respiration (energy release from glucose) would stop, so the cell would not get enough **ATP**. Most cell activities would stop and the cell would soon **die**.

13 **Which phenomenon inhibits the formation of tumors in the human body? Can plants also develop tumors? Explain.**

💡 SAMPLE ANSWER

Contact inhibition: animal cells stop dividing when they touch neighbouring cells. Cancer cells lose this control and form tumours.

Plant cells do not show contact inhibition; their rigid walls give a different growth pattern. Plants can still get **tumour-like growths (galls)**, for example crown gall caused by bacteria, but these do not spread through the body like cancer, because plant cells are held in place by cell walls.

14 **The cell membrane is made up of proteins and lipids. Which cell organelles help in the synthesis of cell membrane? Write the path of these compounds from their site of synthesis to the cell membrane and show this through a labelled diagram.**

✔ ANSWER

Organelles: ribosomes on the **RER** (make proteins), **SER** (makes lipids), **Golgi apparatus** (modifies and packs them).

Path: RER/SER → vesicles → **Golgi apparatus** → vesicles → fuse with the **cell membrane**.

Diagram (draw like Fig. 2.13): nucleus → RER with ribosomes and SER → small vesicles → stack of Golgi sacs → vesicles → plasma membrane; join them with arrows.

15 **What would happen if gametes are formed by mitotic divisions?**

✔ ANSWER

Gametes would have the **full number of chromosomes** instead of half.

At fertilisation the chromosome number would **double** in every generation, causing serious genetic problems. Also, there would be much less variation among offspring.

16 (i) **Deepa preserved her extra amla and lemons as pickles, murabbas and sharbat by adding salt, sugar or jaggery. Which scientific concept has she applied?**

✔ ANSWER

Osmosis – a high concentration of salt or sugar (preservation by salting and sugaring).

16 (ii) **How does the addition of high concentrations of salt and sugar create an environment that prevents the growth of spoilage-causing bacteria and fungi?**

✔ ANSWER

Strong salt or sugar solution is **hypertonic** to microbes. Water moves out of the bacteria and fungi by osmosis; they shrink, cannot grow or multiply, and die. So the food does not spoil.

16 (iii) **Suggest a healthy recipe of this kind for food preservation.**

💡 SAMPLE ANSWER

Amla murabba with jaggery: boil amla pieces lightly, cook them in thick jaggery syrup with a little cardamom, cool and store in a clean, dry glass jar.

(Or lemon pickle: lemon pieces with salt, turmeric and spices, kept in the sun for some days.)

16 (iv) **What are the scientific values addressed in this case?**

💡 SAMPLE ANSWER

- Using scientific knowledge (osmosis) in daily life.
- Reducing food wastage and post-harvest loss.
- Food security, self-reliance and higher income (entrepreneurship).
- Sustainable use of resources and support to the local economy.

⭐ The Quest Continues ...

1 **What is the future of the development of synthetic cells using non-living chemicals?**

💡 SAMPLE ANSWER

Scientists have so far made only synthetic DNA and put it into an existing cell (Venter, 2010). In future they may build simple cells from chemicals. These could make medicines, vaccines and biofuels, clean pollution, and help us understand how life began.

2 **If a synthetic cell is developed, what may be the related ethical issues?**

💡 **SAMPLE ANSWER**

- Safety: it could escape and harm people, crops or nature.
- Misuse, for example to make harmful germs.
- Questions about "creating life" and who owns or controls it.
- Need for strict rules and fair use for everyone.

🎉 **Chapter 2 done! Revise the answers once more tomorrow.**